Diagnostic Exponential Rate Attenuation in Spatial–Temporal Earthquake Models

Stability, identification and predictive evaluation of point-process models

Plain-language companion - diagnostic rate-attenuation and sensitivity-controls edition, 13 November 2026

Includes component-ablation, trend-control, oracle-decomposition and contrast-specific minimum-detectable-effect summaries; sealed authorities unchanged and an additive referee-remediation supplement.

Can a simple statistical rate-reduction term improve forecasts after an earthquake, once the competing model can adjust its background and triggering?

This guide explains a diagnostic comparison of point-process models. The tested attenuation is an isotropic rule for reducing a forecast rate. It does not calculate directional Coulomb stress changes, receiver-fault loading or rate-and-state friction. The earthquake catalogue records event times, locations and magnitudes, not a physical inhibition label.

The principal finding

The tested attenuation additions do not improve the observed held-out comparisons over their matched excitation procedures. A training fit that selects zero attenuation returns the exact comparison model: its held-out contrast is then exactly zero by nesting. This is not a power calculation or a test that physical stress shadows are absent.

Mathematical validity and predictive value are separate

The monograph proves restricted stability and identification results, examines nuisance fitting, and evaluates predictions. A valid mathematical law need not produce useful finite-catalogue learning or a better forecast. The new moderate synthetic regime has a small raw average gain relative to its Monte Carlo error.

How to read this guide

Sections 2-4 introduce the ingredients, memory and scoring. Sections 5-6 explain the completed synthetic and regional comparisons. Sections 7-8 explain interpretation and locate the sources. Chapter 6 of the monograph gives the common study-scope and inference conventions.

Internal independent computational audits and saved-output artifact checks are distinguished from external academic peer review. These historical research comparisons do not establish an operational forecasting service.

02 The ingredients

What goes into a model

Here, a model is a recipe for assigning a changing rate of earthquake occurrence across time and, in the spatial studies, across a region. It gives a distribution of possible activity rather than a single certain date and location for the next earthquake.

Background

Activity not assigned to earlier recorded events

Triggering

Earlier events raise the modelled rate

Inhibition

A proposed reduction in part of the rate

Different models combine these ingredients in different ways.

Background activity

This is the part of the model's event rate that is not attributed to triggering by recorded earlier events. It may change over time and place. Calling an event a background event is a modelling description; it is not a direct observation of its geological cause.

Triggering and diagnostic attenuation

Triggering raises a modelled rate after earlier events. Exponential attenuation multiplies a specified part of that rate by a number no greater than one. One tested version attenuates only the background contribution; another attenuates the total rate. The term inhibition is a statistical model label here, not a recovered physical mechanism.

What spatial kernels represent

The spatial models distinguish a regional background pattern from the offspring pattern associated with earlier earthquakes. The original local attenuation gives eligible parents equal weights. Magnitude-weighted and rupture-scaled extensions give larger parents stronger or wider effects, but remain isotropic point-kernel diagnostics.

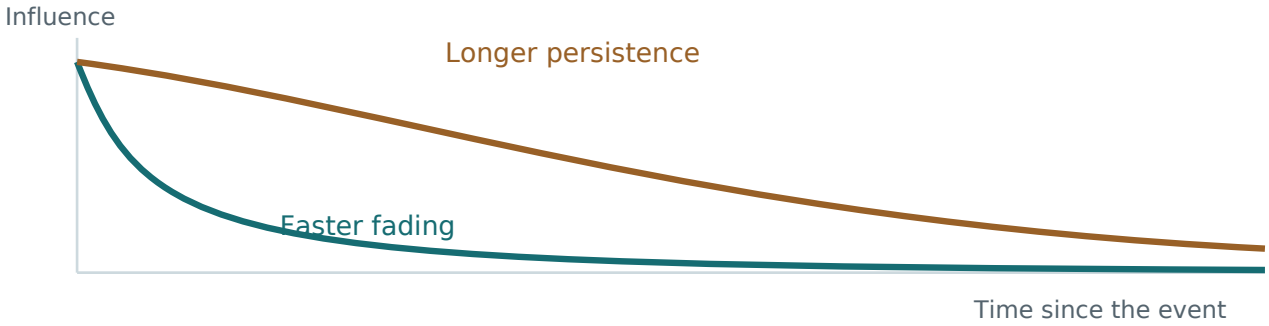
Physical Coulomb models can produce positive and negative directional stress lobes and depend on rupture and receiver-fault geometry. Rate-and-state models connect stress changes to fault response. Those mechanisms motivate scientific context; they are not implemented by the simple isotropic attenuation rules tested here.

A separately executed constant-background fit links the stability model to the data. It uses the same intensity family as an earlier sensitivity comparison, with a different fitting recipe. Inhibition loses in the early and late periods and is switched off in the middle. The theorem does not establish practical equilibration over the catalogue period.

03 The mathematical questions

Memory and what can be learned

Memory means that the influence of an earlier event fades over time. A short memory concentrates influence soon after an event. A longer memory allows it to persist. It is a feature of the model, not an extra quantity directly recorded in the catalogue.



Schematic only. These are not fitted curves or earthquake observations.

Many simple fading effects can represent a complex memory

Some computations combine a finite collection of simple fading components, each with its own speed. This gives the computer a compact state to update as events arrive. It can approximate a more complicated memory law over a chosen range, but a finite approximation is not the same mathematical object as an exact long-memory law.

Stability has a precise and limited meaning

For the autonomous finite-mode model, the results require constant loading, fewer than one directly triggered event per event on average, and suitable conditions on the response and event properties. Under those assumptions, the theory controls expected activity. That does not mean every simulated sequence is small or that all measures of variability are finite. The manuscript shows why first-moment results - results about averages - can coexist with difficulties involving second moments, which concern variability.

Some settings cannot be distinguished

Imagine turning the strength of a model's inhibitory effect to zero. Changing its proposed range or duration then makes no difference to the forecast. The catalogue cannot identify those unused settings from that forecast. A reported range at zero strength is a convention, not a discovered physical length scale.

Even away from zero, two effects can compete to explain the same pattern. A changing background can make it harder to learn an inhibition effect. Knowing that parameters are distinguishable in an ideal mathematical law does not guarantee accurate estimates from the particular catalogue available.

04 Fit, prediction and calibration

How the comparisons work

Fit first, then test on later observations

The model's adjustable settings are fitted using a training history. A predictive comparison then scores observations outside that fitting period under the declared rules. The relevant comparison uses the same events, time window and region for both models.



Illustrative split. Window lengths and study designs vary.

A score rewards useful probabilities

The research uses log scores: a way of evaluating the probabilities or rates a model assigns to what actually happens. A model cannot win simply by announcing high activity everywhere. Its score also accounts for the activity it predicted across the full observation window. Higher scores are better within the same scoring comparison.

Scores can be separated into contributions for timing, location and magnitude. A model may improve where it places activity while making the timing worse. The overall result and the component results therefore need to be read together.

Calibration asks a different question

A forecast that assigns a one-in-five chance to an event should, across suitable repeated comparisons, see that event about one time in five. That is the idea of calibration. In this research, simulated futures are compared with the actual future to look for discrepancies in counts, event spacing and locations.

Winning does not mean passing every test

One model can score better than another while both give poor descriptions of important features of the data. The benchmark's score advantage is not a calibration certificate.

A log-score difference reported in nats is a difference on a logarithmic scale. It is not a percentage improvement or a number of earthquakes correctly predicted.

05 Experiments with known rules

What the controlled studies show

Simulations supply a known generating rule. The outcome is the performance of a particular training-and-selection procedure; a positive generating strength does not guarantee a recoverable signal.

Complete joint fits remove the broad, long-lasting advantage

All 80 original histories and 400 procedure outcomes are complete. Both arms fit their temporal background and triggering parameters on training data. Broad/long immigrant attenuation averages about -10.4 nats per history, with Monte Carlo standard error 8.6 and 10/20 wins. Broad/long total attenuation averages about -1.3 nats, with standard error 2.6 and 8/20 wins. The test windows contain only about 17 and 13 events on average, respectively.

A flexible comparison model changes the answer

Refitting the null background rate removes most of the large gains seen against a fixed null. Also fitting its branching ratio reduces the broad/long total-attenuation residual from about +9.7 to +5.6 nats, with zero branching selected in all 20 histories. Jointly fitting the temporal backbone reverses the residual. Wins by a saved procedure with zero attenuation are comparisons between fitted floors, not detections.

A moderate regime retains a less sparse test window

Twenty fresh histories reject about 27% of proposals and retain about 95 test events on average. The raw mean gain is about +0.29 nats, with Monte Carlo standard error 2.78 and 13/20 wins. A later training-BIC-gated sensitivity, applied to frozen fits, gives a mean of about +2.49, median zero and standard error 1.44, with 9/20 wins and positive attenuation in 12/20 histories.

Thirteen moderate-control fits reach a search boundary. Eighteen do not pass the stricter inherited training score-and-gradient comparison, although they meet the declared score-doubling rule and their held-out scores pass the separate numerical check. These diagnostics are reported with both rows. The narrow/short immigrant controls thin almost nothing; their positive generating strength is not an informative realized suppression signal.

Sample Monte Carlo errors summarize variation across twenty independent simulated histories. Wilson intervals summarize the win fraction, not uncertainty in the mean. Regimes and repeated procedures are not pooled into independent replications.

06 How much could the procedure recover?

Oracle ceiling and detection scales

Separate the generating effect from nuisance estimation

The oracle ceiling separates the exact generating attenuation effect from nuisance adaptation. The full true generating law, including the true background and triggering backbone, beats the fixed true-backbone null by about 16.4 nats and the intercept-refitted true backbone by about 14.5 nats. Against the jointly fitted five-parameter null it gains about 18.1 nats, larger by about 3.6 nats because the uninhibited comparator itself benefits from finite-sample estimation.

Fixing the true attenuation on a fitted backbone yields only about 3.1 nats, and refitting the backbone at true attenuation yields about 2.6 nats. The fitted procedure captures about 0.3 nats, so estimation loss of about 17.8 nats erases almost all of the oracle advantage on these 95-event histories; fitted attenuation strength has mean about 0.93 and range 0.04 to 4.0 against the generating 0.4.

Contrast-specific 90-day detection scales are about 9.2 nats for New Zealand shared attenuation, 3.8 nats for California LI+ total attenuation, 48.3 nats for California EP+ minus EP, 43.3 nats for California OC minus EP, and 47.9 nats for the three-fold Japan OC minus EP total (16.0 nats per fold). A detection scale is not a significance threshold for a dependent, adaptively reused archive.

07 New Zealand, California and Japan

What the regional studies show

New Zealand: compare components and dependence scales

Shared attenuation loses about 17.6 nats in total, with about 94% of the deficit in the first test period. Fixing the background trend at zero reduces the three-fold deficit to about 11.4 nats (8.5 in the first period), and fixing it at the null trend reduces it to about 5.0 nats (4.6 in the first period), below the 9.2-nat 90-day detection scale. Trend and depletion trade off in the fitted event scores. The attenuated model has lower held-out integrated exposure and loses chiefly at event times; the record does not support a simple monotone-stress or rate-overprediction account.

Paired calendar-block summaries retain event and exposure contributions together. Results are shown at 90-, 180- and 365-day scales. The second-period interval spans zero at each scale; the third-period interval changes with scale. Long-memory dependence and exchangeability are not established, so the intervals are descriptive sensitivity summaries and the sign-flip references are nominal, not calibrated significance tests.

California: a stronger excitation comparison

The original unweighted benchmark, EP, beats the background-weighted recipe, OC, in each primary California fold, mainly through location; a separate episode backtest has one reversal. Both original spatial procedures show conditional simulation discrepancies. Better scores alone do not establish calibration.

The stronger ETAS procedure gains about 69.6 nats over archived EP while changing four ingredients together: magnitude-dependent offspring distance, immigrant-weighted background, joint fitting, and full-plane normalization with regional leakage. All three fitted spatial scales reach the 2 km lower search bound. Relaxing that bound selects interior scales of about 0.82 to 0.84 km and raises the gain to about 77.1 nats, or 78.6 nats with the accepted weights held fixed. The gain does not isolate one causal ingredient.

Zero attenuation gives an exact nested tie

The corrected final optimization includes positive-strength initial states. It selects zero attenuation in all three California folds. The scored states then coincide exactly with the matched excitation baseline, so the held-out contrasts are exactly zero. This procedure result does not measure detection power or rule out a physical stress-shadow mechanism.

Japan and the smaller California contrasts

The timing advantage of the original California recipe becomes a loss in Japan; fixing background trend reduces but does not remove that deficit. The original California total-attenuation losses are small, with block summaries that vary across scales and nominal reference tests. Read fold consistency and score components alongside the intervals, rather than assigning a calibrated significance label.

08 The boundaries of the evidence

Interpreting the evidence

A diagnostic comparison is not a physical absence test

The tested attenuation recipes are fitted statistical extensions. Their outcomes distinguish these procedures and comparators on the supplied histories. Physical Coulomb stress and rate-and-state friction require different directional and fault-response structure; selecting zero in the present family does not adjudicate those mechanisms.

Looking back is different from issuing a forecast

Using only earlier data when fitting is an important safeguard. It does not turn a retrospectively chosen set of episodes into a genuinely prospective study. Overlapping training histories and related earthquake sequences also limit how many independent pieces of evidence the comparisons supply.

Readers can now rebuild the study inputs

Original provider records and selection rules are now available for the New Zealand reconstruction, California and Japan. Replays rebuild their study catalogues from those records. A fresh run of the frozen New Zealand comparison also reproduced its fitted parameters, event-level scores and numerical controls exactly. This makes those calculations inspectable and repeatable; reused observations remain part of model development. Later CPU runs also recalculated the declared historical comparison scores from the original saved model settings. Their records distinguish rebuilt numerical inputs from exact saved inputs. These checks repeat the scoring, not the original model searches, and do not turn reused observations into new test data.

Better scores do not settle calibration

The completed study compares OC and EP with 999 simulated futures at each of three forecast origins. Combined checks using continuous timing and spatial statistics flag discrepancies for both models in every origin. EP can score better while still giving an inadequate description of some features. These checks hold fitted parameters fixed; they do not include uncertainty about those parameters.

What the supplied records allow

The supplied inputs, fitted states and source epochs let readers distinguish fresh computations, saved-state score replays and historical records. Internal independent computational audits state which arithmetic or raw calculations were checked. A passing hash or saved-output artifact check is not external peer review.

The F continuation has been removed because required original fitted states could not be recovered. Its older sources and numerical records remain archived; the supported earlier nuisance study is retained.

09 Finding your way through the manuscript

A small glossary and reading map

Catalogue	The recorded times, locations and magnitudes of selected earthquakes.
Parameter	An adjustable model setting, such as the strength or duration of an effect.
Inhibition	Here, a diagnostic exponential attenuation term that reduces a specified model rate; not an identified physical stress mechanism.
Identifiability	Whether different settings could be distinguished from the model's observable law, in principle.
Null control	A simulated case without the effect the procedure is supposed to detect.
Oracle calibration	A simulation-based reference using the known generating law.
Held-out data	Observations outside a particular fitting step; repeated inspection can still make them part of model development.
Zero-strength boundary	The setting where inhibition is switched off and its unused range and memory cannot be identified.
Calibration	Agreement between a forecast's probability statements and repeated outcomes under the declared comparison.
OC, EP, LI and LI+	OC uses a regional background map to weight where triggered activity is placed; EP omits that weighting and uses its own distance-decay rule. LI adds local background suppression to OC; LI+ gives larger events stronger suppressing weights.

Where to read more

The model and its mathematical limits	Chapters 2-5
Study design and controlled detection	Chapters 6-7
Historical temporal and magnitude comparisons	Chapters 8-9
Spatial models, regional transfer and backtests	Chapters 10-11
Calibration and local inhibition	Chapters 12-13
Overall interpretation and remaining questions	Chapter 14

Read results with their comparators and assumptions. Start with docs/data.md and reproduction/README.md; the selected-state reader and docs/cpu_reproduction.md document executed Linux routes. The amended reader at experiments/s04b_refined_cpu_reader/README.md supplies an audited Linux library and finite local limits. Its outputs require separate review; the earlier CPU acceptance covers a different route. Readers can inspect the evidence, report errors and propose revisions.

Terms follow the diagnostic monograph. Chapter 6 states the shared study scope; current records distinguish fresh fits, score replays and historical artifacts.